

Solar Sector: India's Next Employment Hub

While solar energy production has the potential to end the power woes of a country, the added advantage is that it also creates a plethora of new job opportunities for its citizens. Find out the solar scenario in India, job opportunities available, skill set in demand and, of course, the pay package



ABHISHEK MUTHA

The four most useful natural energy resources that would probably remain always available on the Earth are wind, waves, geothermal heat and solar power. Among these, solar power is one of the most cost-effective sources of tapping energy.

Though initial installation and maintenance cost of the solar panels is on the higher side, these are an absolutely green way to generate electricity—free of smoke, gas and chemicals. Higher demand, cheaper materials, lower installation and maintenance costs, better policies and improved standards will surely make solar energy affordable in the near future. Also, initiatives like the Jawaharlal Nehru

National Solar Mission (JNNISM) by the government of India, will lead to strengthening of the manufacturing capabilities and technology advancement, creating many more jobs.

Solar scenario in India

Meeting the increasing demand for electricity and expanding electricity access to remote villages are the two major challenges before India. Moreover, the Indian power sector is highly dependent on coal as fuel (see Fig. 1). "Solar is one sure shot initiative to address this, with coal becoming a rare commodity and carbon footprint and green initiative pushing us to embrace solar energy much more than any other technology," says Prakash Nayak, chairman, IET India Power Engineers Panel.

Sriram Ramakrishnan, CEO, Consul Consolidated, adds, "In India, as there is an increasing gap between supply and demand of power, the solar sector is poised for explosive growth over the next decade." According to a report, India faces a power deficit of about 11 per cent in the overall demand and 12 per cent in peak demand (see Fig. 2). Also, with an expanding economy, the demand for power is growing at the rate of approximately 6 per cent every year.

"Solar power is the major key to bridge this increasing demand-supply gap in energy," says Abhay Singh, director-HR, Applied Materials India.

Though energy conservation is a major concern in the present situation, India can take full advantage of its geographical position as it is a sun-rich country with tropic of cancer passing through it. "On an average, we get more than 300 days of good solar radiation per year as compared to a cold country like Germany where solar photovoltaic systems have made economic sense," says Ganesh Shankar, managing director, FluxGen Engineering Technologies.

"India is fortunate to have over 3000 hours of sunshine annually, which can be utilised as an alternative source of energy with the help of solar photovoltaic (PV) or concentrated solar thermal power technology," says Nayak.

One of the major developments in the Indian solar sector is the JNNISM initiative, which basically aims at generating 20 GW of solar power by year 2020. Prior to JNNISM, for solar PV, the focus of the government was primarily